

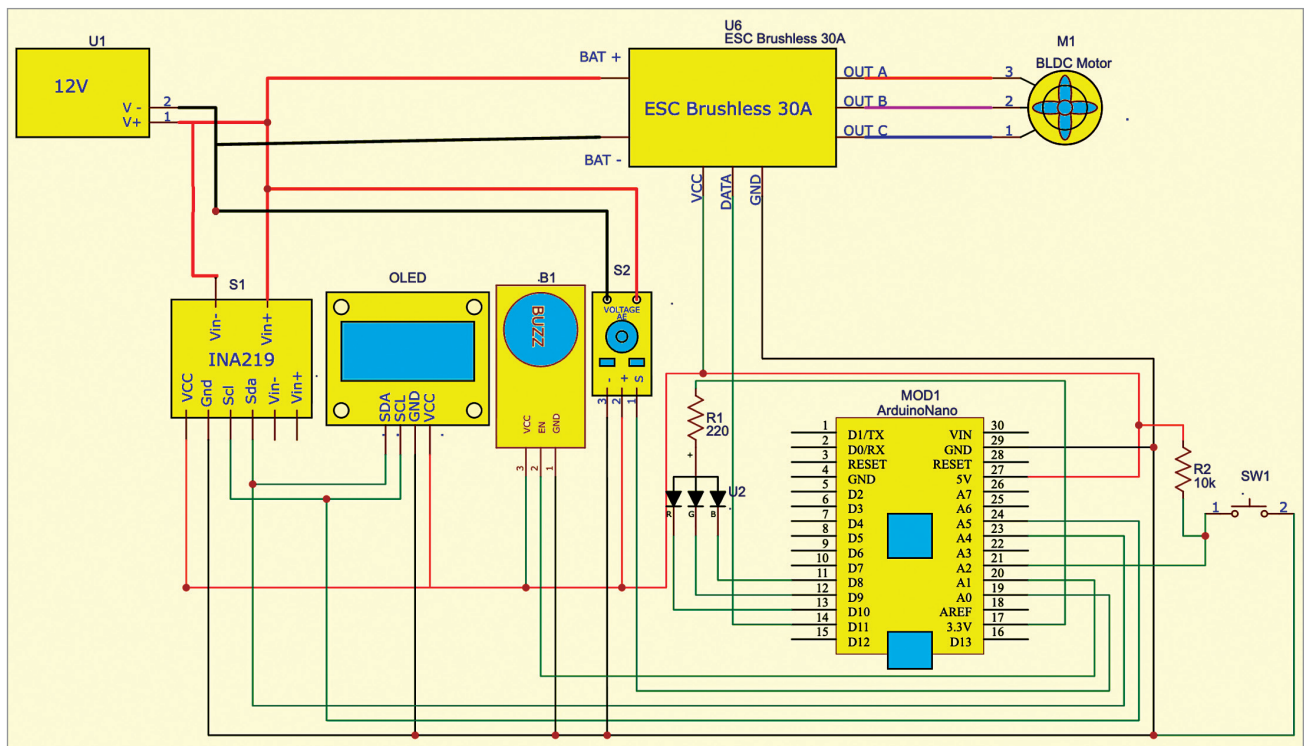


Fig. 3: Circuit diagram

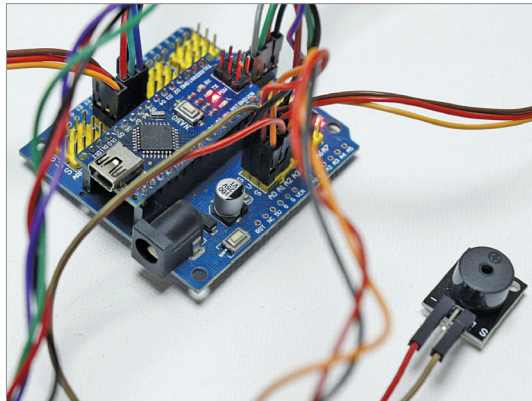


Fig. 4: Arduino Nano connection with the expansion board

Excessive current points to an overloaded or mechanically stressed motor, while low voltage indicates a weak or unsafe battery condition. Fig. 1 shows the author's prototype.

Using carefully defined thresholds, the system accurately classifies each fault before the drone take-off stage. If an unsafe condition is detected, the system immediately shuts down the motor and provides clear feedback via an OLED display, RGB LED indicators, and a buzzer. Instead of guessing or interpreting raw values, the user receives a

direct explanation of the issue.

The smart drone safety system is a standalone safety module designed to prevent damage to motors, ESCs, batteries, and frames. It reduces trial-and-error troubleshooting and serves as a quick diagnostic tool for technicians. Additionally, it acts as an educational platform for students and drone builders by demonstrat-

ing real motor behaviour through measurement-based analysis.

This system is simple, affordable, and practical. It can be used during assembly, maintenance, or routine pre-flight checks, improving reliability and promoting safer drone operation. The system also supports future expansion, including multi-motor testing, data logging for long-term motor health tracking, and wireless monitoring integration. The components required to build this system are listed in the Bill of Materials Table 1.

Software coding

The Arduino IDE platform is used to write, compile, and upload code to the Arduino board. It supports C/C++-based programming, provides a simple user interface, includes built-in libraries, offers serial monitoring, and ensures cross-platform compatibility. This environment makes embedded system development accessible for both beginners and professionals.

The INA219 sensor is used to measure current, while the SSD1306 OLED display is used for data visualisation. To interface these devices, the libraries `Adafruit_INA219.h`, `Adafruit_GFX.h`, and `Adafruit_SSD1306.h` must be installed using the Library Manager.

After installing the required libraries, the pins for the sensors, RGB LED, and ESC PWM signal are defined in the code. The Servo library generates the PWM signal required to control the ESC. Fig. 2 shows a snippet of the source code. The program displays the system status using the RGB LED, and the